

Evaluation to digital filtering to obtain Sagnac shift in open loop-IFOG

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Abstract

Purpose – The purpose of this paper is to find the best way to obtain the Sagnac phase shift in the output signal of open loop interferometric fiber optic gyroscopes (IFOGs). Also, the utilized digital filtering based on FIR kaiser window for implementing the digital signal processing part is evaluated.

Design/methodology/approach – The approach is based on implementing four kaiser FIR filters, the coefficients of which have been obtained from SPtool. They were simulated with SPtool in the Matlab 7.1.

Findings – The results show that the chosen computational method has reliable accuracy. On the other hand, it could require low-computational effort, and it is a simple way which is important for the signal processors.

Research limitations/implications – The limitation in this paper is that the designed filters have high order and they require much time; therefore, a high-speed device is needed. For solving this problem, it is proposed to perform some estimation by experiments.

Practical implications – IFOGs are used in aircraft, missiles, and new civil fields such as automobile navigation, antenna stabilization, crane control, unmanned vehicle control, wind, and renewable energy platform stabilization.

Originality/value – There is no other paper which has explained mathematics of IFOG implementation in the signal processing part as completely as is done here.

Keywords Signal processing, Rotational motion, Sensors

Paper type Research paper

I. Introduction

Angular rotation is one of the parameters that needs to be accurately measured in order to describe the position of an object. Traditionally, the angular momentum of a spinning rotor was used to determine the angular rate or displacement. These devices are susceptible to damage from shock and vibration, exhibit cross-axis acceleration sensitivity, and for the lower cost versions, have reliability problems (Asami *et al.*, 1998). Several alternative technologies are being developed including Coriolis effect rate sensors (Asami *et al.*, 1998; Kelley *et al.*, 1992). Discovered in 1913, the Sagnac effect found its first practical application in the ring laser gyroscope (RLG), now used extensively in commercial inertial navigation systems for aircraft. But, since this implementation requires high vacuum and precision mirror technology, cost has been a factor limiting its application to more mundane tasks. The physical principal of RLG operation is analogous to the Doppler effect, but in this instance, it involves determination of the phase shift between two counter-propagating light beams. For a RLG, this occurs in an evacuated mirrored cavity, but in the interferometric fiber optic gyroscopes (IFOGs), the same effect can be obtained in

a fiber coil, eliminating the high voltage and high vacuum; so, making low-cost inertial rotation sensors practical (Kelley *et al.*, 1992; Merlo *et al.*, 2000). The performance of fiber optic gyros (FOG) is mainly characterized by bias stability, scale factor linearity, and random noise component termed angle random walk (Seidel, 2004). Within the scope of FOG technology are bias stabilities ranging from 1 °/s to 0.001 °/h; scale factor linearity about 10,000 ppm and random walk from 200 to 0.001 (°/h)/√Hz. The advantages of the IFOG are attributed to their all solid-state construction and their limited number of components; so, lower cost (Merlo *et al.*, 2000).

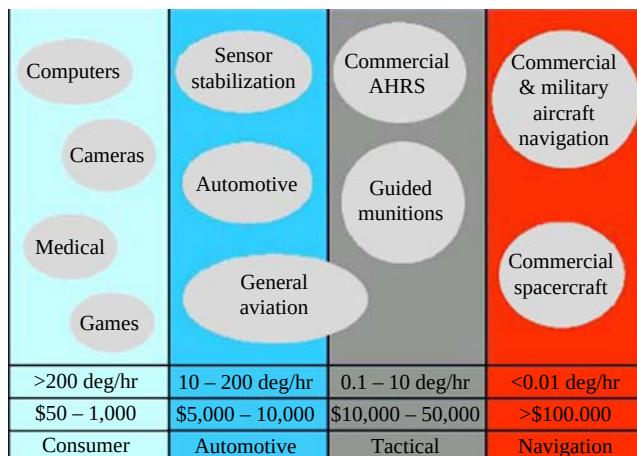
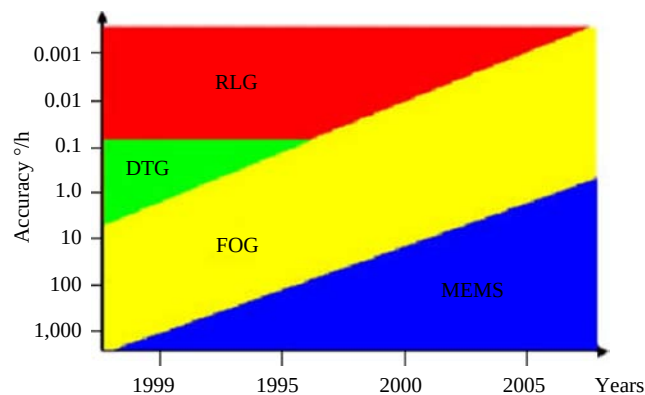
II. Optical gyros comparison

Modern inertial navigation systems require sensors that are highly reliable and have a wide dynamic range. These objectives are not easily achieved with mechanical gyroscopes. The products of RLG have already existed, offer major advantages over their mechanical predecessors in the areas of reliability, cost, power consumption, startup time, shelf life, and *g*-sensitivity (Lefevre, 1997). IFOGs share the advantages of the RLG and are being pursued primarily because they offer the potential of further improvements in the areas of reliability, cost, and weight (Bennett *et al.*, 1998). With considering above, however, accuracy is good but it is caused to grow up cost of the gyro. There is one point; we do not need high-precision gyro in all applications. For instance, in automotive application, it needs utilizing IFOG with accuracy between 10 and 200 °/h. This matter could be one important parameter to design an appropriate IFOG. The variety of IFOGs and their accuracy are shown in Figures 1 and 2.

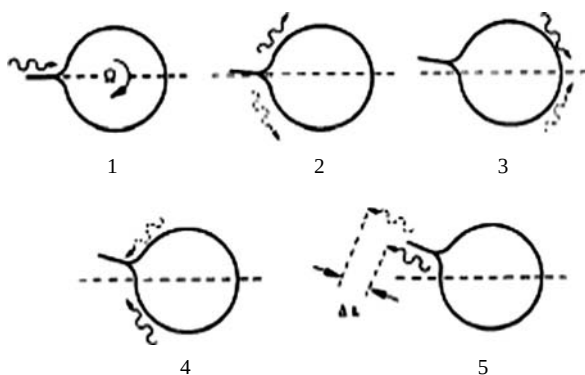
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Figure 1 Chart of accuracy and cost gyroscopesSource: Shaikh *et al.* (2005)**Figure 2** Evolution of gyro technologiesSource: Shaikh *et al.* (2005)

All three kinds of optical gyros, the IFOG, the resonator FOG (RFOG), and the RLG are based on the Sagnac effects, as shown in Figure 3. This phenomenon predicts that two optical beams traveling in opposite directions around the same closed optical path will experience path length differences that are proportional to the rotation rate of the setup (Bennett *et al.*, 1998; Oho *et al.*, 1995).

Figure 3 Sagnac effect

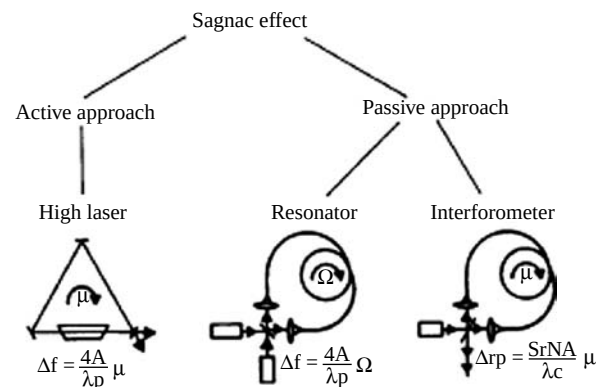
Source: Avanaki (2006)

The various optical gyro technologies are distinguished by different approaches in measuring this rotation induced path length difference. The actual path length difference due to rotation is quite small. For example, gyro used in aircraft navigation must detect rotation rates below 0.01 °/h. This implies sensing path length changes of about one part in 10^{16} , which corresponds to absolute length changes on the order of a nuclear dimension. In achieving this precision, the optical gyro has clearly established itself as one of the most exacting instruments in existence. The means of implementing the measurement of path length difference are quite different among the RLG, IFOG, and RFOG, as shown in Figure 4. Both the RLG and the RFOG use an external light source to launch light into the passive fiber loop. The IFOG requires a broadband light source to eliminate many error sources, while the RFOG, whose operation most closely resembles the RLG, requires an extremely narrowband light source. For the RLG, narrow band gas laser with its active gain medium is an integral part of the sensing cavity. The RLGs are divided to two categories: “active” and “passive.” The RFOG and RLG share the same scale factor equation and their path length differences are sensed by measuring the difference between two cavity resonant frequencies. In the IFOG, the rotation rate sensing is achieved through direct measurement (open loop) or nulling (close loop) of the optical phase differences of the accomplished by applying a phase shift equal and opposite to the rotation-induced Sagnac phase shift (Merlo *et al.*, 2000; Avanaki, 2006).

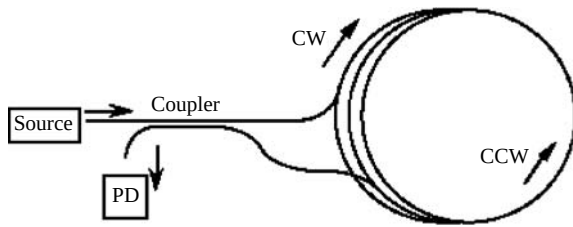
III. Principle of operation

The principle of operation of an IFOG is quite simple. It is based on the phenomenon that a circuitual system has different optical path lengths in the two propagation directions when it is rotated. The difference between counter propagating beams provides amount of the rotation (Figure 5).

The main structure include of a laser diode, usually super luminescent diode (SLD) or rare-earth doped fiber, with an improved wavelength stability, a coupler to split light into clockwise and counterclockwise directions, a length of optical fiber wound on a coil as the rotation sensing loop, and a photo detector to convert optical information into electrical signals for further processing. Light source temperature is controlled by using a thermo electric cooler (TEC) and type of the

Figure 4 Rotation measurement in three commonly known optical gyro technologies: RLG, RFOG, and IFOG

Source: Avanaki (2006)

Figure 5 The basic scheme of the IFOGSource: Merlo *et al.* (2000)

detector is a PINFET module having high sensitivity and hybrid trans-impedance amplifier. To make a more accurate IFOG, it is necessary to have identical optical path, a polarizer and a second coupler. By this way, the created IFOG is called “minimum configuration” shown in Figure 6 (Perlmutter, 1994; Emge *et al.*, 1997; Shidlovski, 2004).

Phase modulator added to the loop-to-phase modulation. The combined signal is passed via the fiber coil and couplers, received by the detecting means. Light detection means convert combined beam in an electric current. An electric amplifier is joined to the detector for changing current to an electric voltage. Alternating current voltage amplitude controller means is joined to the amplifier, and controls the source intensity. Electrical signal processing means is joined to the amplifier and processes the voltage and provides a production signal relative to the angular rotation rate input of the sensing loop (Avanaki, 2006; Nabavi and Nayeri, 2000).

IV. Rotation equations

When the optical ring is rotated with a tangential velocity v , the beam rotating with the ring will have an optical path longer than the counter-rotating beam by a distance ΔL given by:

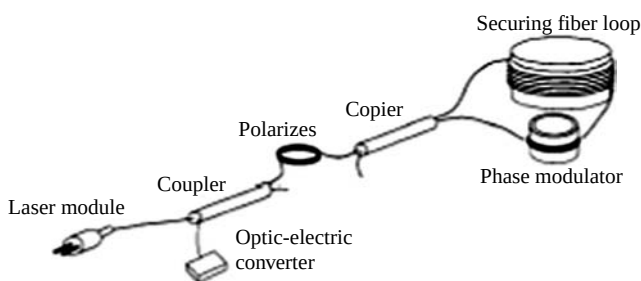
$$\Delta L = \frac{4\pi Rv}{c} \quad (1)$$

where, R is the ring radius and c is the speed of light in the vacuum.

For a monochromatic light of wavelength λ , this change in optical path length, results in the Sagnac phase difference which is given by equation (2):

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta L = \frac{8\pi^2 Rv}{\lambda c} \quad (2)$$

The phase difference between two beams, after passing around the ring with area A and rotating with an angular velocity Ω , makes the phase difference is given by:

Figure 6 Minimum configuration IFOGSource: Oho *et al.* (1995)

$$\Delta\phi = \frac{8\pi A\Omega}{\lambda c} \quad (3)$$

It is important to note that, the resultant phase shift is independent of the medium and of the exact shape of the loop. This compliant property of the sensor portion of a IFOG is an advantage when it is required to design IFOG to fit the volume constraints in specific applications.

The resultant phase shift can be increased by additional loops (turns of fiber). If it is wound N turns of the fiber coil, the resultant phase shift becomes:

$$\Delta\phi = \frac{8\pi A\Omega N}{\lambda c} \quad (4)$$

Alternatively, we can express the resultant phase shift in terms of coil diameter and fiber length by noting that:

$$A = \frac{\pi D^2}{4} \quad (5)$$

and:

$$L = N\pi D \quad (6)$$

So, the Sagnac phase shift can be rewritten as:

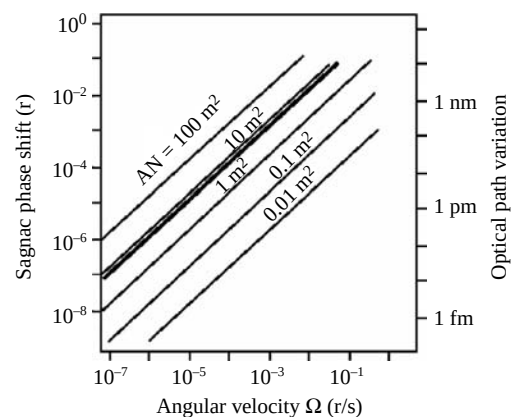
$$\Delta\phi = \frac{2\pi LD}{\lambda c} \Omega \quad (7)$$

The Sagnac phase shift as a function of the angular velocity is shown in Figure 7 for different values of the total enclosed area AN (Seidel, 2004; Oho *et al.*, 1995; Avanaki, 2006). With a typical IFOG (200 m coil length, 10 cm diameter coil), the measurement of the earth angular rotation $\Omega = 15^\circ/\text{h} = 0.73 \mu\text{r/s}$ requires to detect a phase difference $\Delta\Omega = 36 \mu\text{r}$, corresponding to an optical path difference of the order of 10^{-12} m.

The output signal for such configuration will be:

$$I(t) = I_0(1 + \cos(\Delta\phi)) \quad (8)$$

In which, I_0 is the current obtained when the ring is at the rest.

Figure 7 The Sagnac phase shift as a function of the angular velocity Ω for different values of the total enclosed area AN Note: $\lambda = 850$ nm;Source: Merlo *et al.* (2000)

V. The open-loop configuration with phase modulation

As mentioned above, the major problem of the basic configurations shown in Figure 5, is the output nonlinearity for small phase shift, $\Phi_s \approx 0$ which hinders high-sensitivity measurements of small rotation angles without sign ambiguity. This limitation is overcome by transforming the base band cosine-dependence into a sinusoidal function. Although, for translating the output signal from base band to a carrier at angular frequency ω_{mod} , different solutions have been proposed and demonstrated, nowadays the optical phase modulation technique is commonly used. The typical setup of a practical IFOG in all-fiber technology is shown in Figure 8, where a phase modulator is inserted in the fiber coil, close to a coupler output, so that a different phase delay is cumulated by the counter propagating waves (Lefevre, 1997; Liu and Adams, 1990). The all-fiber version of IFOG has a phase modulator which is constructed by winding and cementing a few fiber turns on a short, hollow piezoceramic tube (PZT). By applying a modulating voltage to the PZT, a radial elastic stress and a consequent optical path length variation due to the elasto-optic effect are generated (Avanaki, 2006).

VI. Modulation equations

In the configuration shown in Figure 8, the CCW and the CW propagating waves experience a phase delay $\phi(t)$ and $\phi(t - \tau)$, respectively. Where $\tau = L/v$ is the radiation transition time in a fiber with an overall length of L . By applying a phase modulation at angular frequency ω_{mod} , the modulation equation will be:

$$\phi_{\text{mod}}(t) = \phi_{\text{mod},0} \cos(\omega_{\text{mod}} t) = \phi_{\text{mod},0} \cos(2\pi f_{\text{mod}} t) \quad (9)$$

Here, $\phi_{\text{mod},0}$ is the amplitude of modulation signal, and f_{mod} is the modulation frequency.

Therefore, CW and CCW waves are modulated as phase shift by $\phi_{\text{mod}}(t)$ and $\phi_{\text{mod}}(t - \tau)$, respectively. So, the modulation phase shift difference is as follow:

$$\Delta\phi_{\text{mod}}(t) = \phi_{\text{mod}}(t) - \phi_{\text{mod}}(t - \tau) \quad (10)$$

Finally, the total phase shift in the output signal will be:

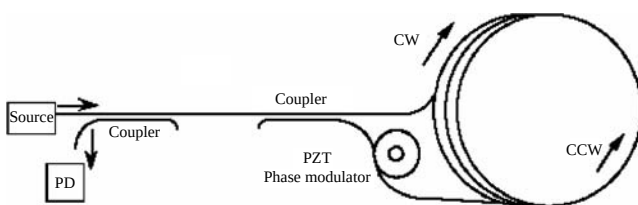
$$\Delta\phi_{\text{total}} = \phi_{\text{ccw}} - \phi_{\text{cw}} = \Delta\phi + \phi_{\text{mod}}(t) - \phi_{\text{mod}}(t - \tau) \quad (11)$$

Now, the output signal will become:

$$I(t) = I_0(1 + \cos(\Delta\phi + \Delta\phi_{\text{mod}}(t))) = I_0(1 + \cos(\Delta\phi_{\text{total}})) \quad (12)$$

Considering the SLD switched by $f_p = 1/2\tau$, with the help of equations (9) and (11), the phase shift becomes:

Figure 8 All-fiber gyroscope with phase modulator



Source: Merlo et al. (2000)

$$\Delta\phi_{\text{mod}}(t) = \Delta\phi + \phi_{\text{mod},0} \cos(2\pi f_{\text{mod}} t) - \phi_{\text{mod},0} \cos(2\pi f_{\text{mod}} (t - \tau)) \quad (13)$$

$$\begin{aligned} \Delta\phi_{\text{mod}}(t) &= \Delta\phi + 2\phi_{\text{mod},0} \sin(\pi f_{\text{mod}} \tau) \\ &\quad \times \sin\left(2\pi f_{\text{mod}} \left(t - \frac{\tau}{2}\right)\right) \\ &= \Delta\phi + 2\phi_{\text{mod},0} \sin\left(\frac{\pi}{2} \frac{f_{\text{mod}}}{f_p}\right) \\ &\quad \times \sin\left(2\pi f_{\text{mod}} t - \frac{\pi}{2} \frac{f_{\text{mod}}}{f_p}\right) \end{aligned} \quad (14)$$

Substituting the amplitude as follow:

$$\phi_{\text{MI}} = 2\phi_{\text{mod},0} \sin\left(\frac{\pi}{2} \frac{f_{\text{mod}}}{f_p}\right) \quad (15)$$

The output current of photodiode becomes:

$$I(t) = I_0 \left[1 + \cos\left(\Delta\phi + \phi_{\text{MI}} \sin\left(2\pi f_{\text{mod}} t - \frac{\pi}{2} \frac{f_{\text{mod}}}{f_p}\right)\right) \right] \quad (16)$$

$$\begin{aligned} I(t) &= I_0 \left[1 + \cos(\Delta\phi) \cos\left(\phi_{\text{MI}} \sin\left(2\pi f_{\text{mod}} t - \frac{\pi}{2} \frac{f_{\text{mod}}}{f_p}\right)\right) \right. \\ &\quad \left. - \sin(\Delta\phi) \sin\left(\phi_{\text{MI}} \sin\left(2\pi f_{\text{mod}} t - \frac{\pi}{2} \frac{f_{\text{mod}}}{f_p}\right)\right) \right] \end{aligned} \quad (17)$$

Figures 9 and 10 show the photo detector response in the domains of time and frequency for both stationary and rotating states of an IFOG.

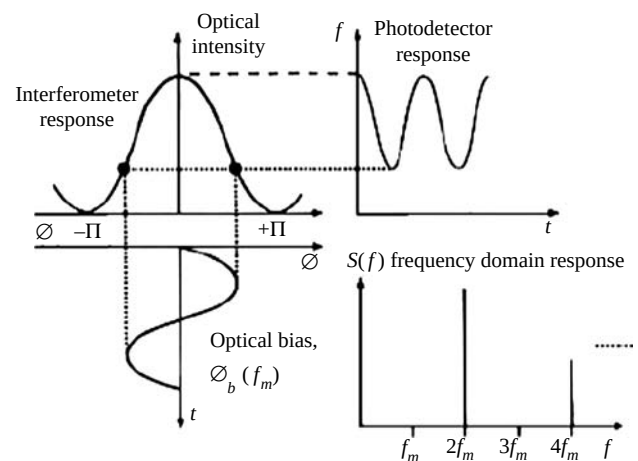
Using the below Bessel equations:

$$\cos(x \sin s) = J_0(x) + 2 \sum_{m=1}^{\infty} J_{2m}(x) \cos(2ms) \quad (18)$$

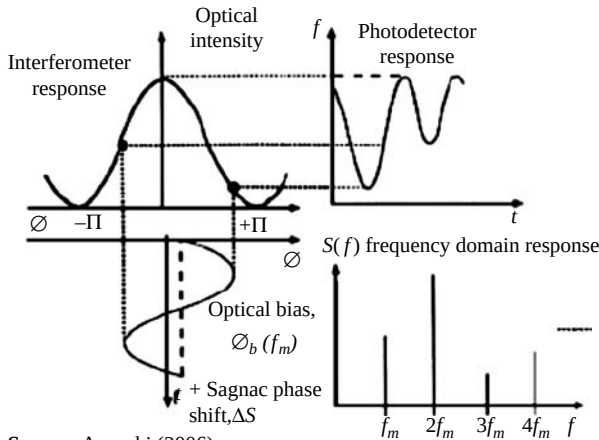
$$\sin(x \sin s) = 2 \sum_{m=1}^{\infty} J_{2m-1}(x) \sin((2m-1)s) \quad (19)$$

The equation (17) can be rewritten in away that one can easily extract the relevant harmonics:

Figure 9 Modulating of the IFOG signal in stationary status



Source: Avanaki (2006)

Figure 10 Modulating of the IFOG signal in rotating status


Source: Avanaki (2006)

$$\begin{aligned}
 I(t) = I_0 & \left[1 + \cos(\Delta\phi) \cdot \left\langle \mathcal{J}_0(\phi_{MI}) + 2 \sum_{m=1}^{\infty} \mathcal{J}_{2m}(\phi_{MI}) \right. \right. \\
 & \times \cos \left(2m \left(2\pi f_{mod} t - \frac{\pi}{2} \frac{f_{mod}}{f_p} \right) \right) \Bigg\rangle \\
 & - \sin(\Delta\phi) \cdot \left\langle 2 \sum_{m=1}^{\infty} \mathcal{J}_{2m-1}(\phi_{MI}) \right. \\
 & \times \cos \left((2m-1) \left(2\pi f_{mod} t - \frac{\pi}{2} \frac{f_{mod}}{f_p} \right) \right) \Bigg\rangle \Bigg]
 \end{aligned} \quad (20)$$

To have an optimum sensitivity, output power, photon noise, and signal to noise ratio, ϕ_{MI} is selected to be as (Seidel, 2004):

$$\phi_{MI}(\text{opt}) = 1.85 \text{ rad} \quad (21)$$

To simplify the equation (20), we put $f_{mod} = f_p$. Using this assumption and the equation (15), we will have:

$$\phi_{MI} = 2\phi_{mod,0} \rightarrow \phi_{mod,0}(\text{opt}) = 0.925 \quad (22)$$

The amplitude of the output signal at f_{mod} , which gives the first harmonic of the output current, i.e. equation (20) will be:

$$I(f_{mod}) = -2C_1 I_0 \sin(\Delta\phi) \mathcal{J}_1 \left(2\phi_{mod,0} \sin \left(\frac{\pi}{2} \frac{f_{mod}}{f_p} \right) \right) \quad (23)$$

$$I(f_p = f_{mod}) = -2C_1 I_0 \sin(\Delta\phi) \mathcal{J}_1(2\phi_{mod,0}) \quad (24)$$

Table I shows all the harmonics generated by photodiode output current.

Now, we can calculate $\Delta\phi$ by H_1 with the below equation:

$$\Delta\phi = -\arcsin \left(\frac{H_1}{2C_1 I_0 \mathcal{J}_1(\phi_{MI})} \right) \quad (25)$$

Table I Harmonics of photodiode output current

Frequency	Amplitude
DC	$H_0 = C_0 I_0 (1 + \cos(\Delta\phi) \mathcal{J}_0(\phi_{MI}))$
f_{mod}	$H_1 = -2C_1 I_0 \sin(\Delta\phi) \mathcal{J}_1(\phi_{MI})$
$2f_{mod}$	$H_2 = 2C_2 I_0 \cos(\Delta\phi) \mathcal{J}_2(\phi_{MI})$
$3f_{mod}$	$H_3 = -2C_3 I_0 \sin(\Delta\phi) \mathcal{J}_3(\phi_{MI})$
$4f_{mod}$	$H_4 = 2C_4 I_0 \cos(\Delta\phi) \mathcal{J}_4(\phi_{MI})$

Note: $C_i = C$, $i = 0, \dots, 4$

$\Delta\phi$ could be also obtained with another approach using H_1 , H_2 , $\mathcal{J}_1$ and $\mathcal{J}_2$:

$$\Delta\phi = -\arctan \left(\frac{H_1 \mathcal{J}_2(\phi_{MI})}{H_2 \mathcal{J}_1(\phi_{MI})} \right) \quad (26)$$

However, the most comfort and reliable method is to use H_1 - H_4 to calculate $\Delta\phi$ with the help of equation (27) and Table I:

$$\frac{2m}{x} \mathcal{J}_m(x) = \mathcal{J}_{m-1}(x) + \mathcal{J}_{m+1}(x) \quad (27)$$

$$\frac{H_2}{H_4} = \frac{\mathcal{J}_2(\phi_{MI})}{\mathcal{J}_4(\phi_{MI})} \quad (28)$$

$$H_1 + H_3 = \frac{4H_2}{\phi_{MI}} \tan(\Delta\phi) \quad (29)$$

$$H_2 + H_4 = \frac{6H_3}{\phi_{MI}} \frac{1}{\tan(\Delta\phi)} \quad (30)$$

Finally, it is obtained:

$$\Delta\phi = \arctan \sqrt{\frac{3H_3(H_1 + H_3)}{2H_2(H_2 + H_4)}} \quad (31)$$

From the hardware point of view, the implementation of these approaches is done by signal processing part such as microcontroller or DSP processor (Merlo *et al.*, 2000; Seidel, 2004; Avanaki, 2006; Nabavi and Nayeri, 2000).

VII. Simulation results

The simulation results of the above-mentioned approach show that this approach can be perfectly be used in signal processing of the IFOGs. At the first step of the simulation, the filters were designed by SPtool in Matlab. The FIR filters were selected for filtering and designing the four requested band pass filters. The inputs for this stage are listed below:

- Input rotation rate = $10^\circ/\text{s}$ (AHRS range).
- Coil length = 500 m.
- Coil diameter = 10 cm.
- Sampling frequency = 10 MHz.
- Time domain = 2 ms.
- Wave length = 1,310 nm.
- Refractive index of optical fiber core = 1.43.
- Coil length transient time = $(500/3 \times 10^8) \times 1.43$.

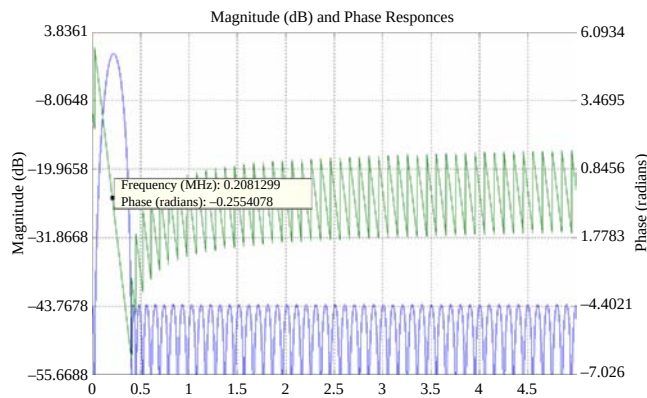
The coil frequency is $1/2\tau = 1/(2L/(c/1.43)) = 0.208 \text{ MHz}$. As mentioned before, this frequency is the PZT vibration and SLD switch frequency, as well. Figures 11-14 show the designed band pass filters, with a same bandwidth of 0.208 MHz. In these figures, F_p and F_s mean pass band and stop band frequencies, respectively. Also, R_p and R_s are ripples in pass and stop bands:

- 1 – Band pass filter. Gain = 1, range (around 0.208 MHz).
- 2 – Band pass filter. Gain = 1, range (around 0.416 MHz).
- 3 – Band pass filter. Gain = 1, range (around 0.624 MHz).
- 4 – Band pass filter. Gain = 1, range (around 0.832 MHz).

To derivate harmonics from output current of IFOG, the implementation of these filters was done using the obtained coefficients and by direct form structure. Figures 15-18 show the filtered signal, H_1 - H_4 .

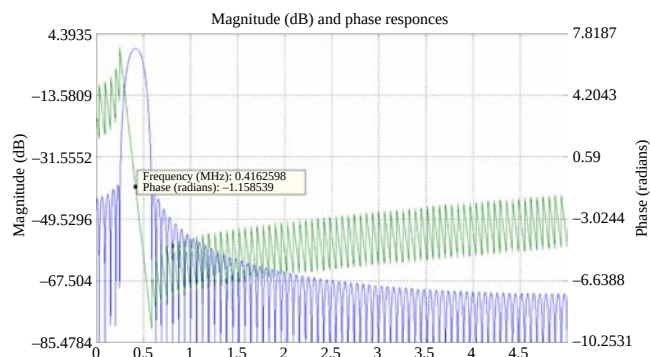
At the next step, Ω was calculated using H_1 - H_4 with the help of equations (7) and (31).

Figure 11 Band pass filter (harmonic 1)



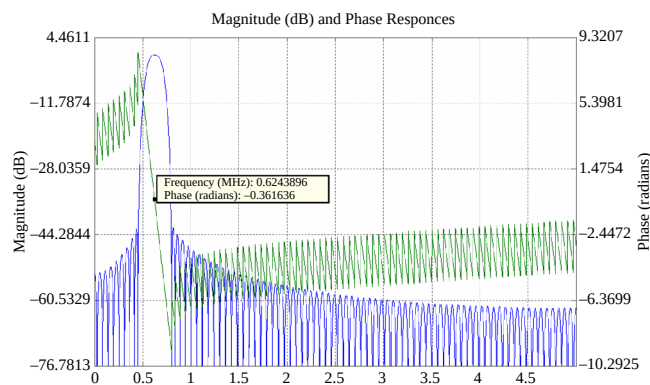
Notes: Kaiser window FIR with order = 100; $F_{p1} = 0.202$ MHz; $F_{p2} = 0.21$ MHz; $F_{s1} = 0.334$ MHz; $F_{s2} = 0.399$ MHz; $R_p = 0.3917$; $R_s = 39.45$; phase delay = -0.255 rad

Figure 12 Band pass filter (harmonic 2)



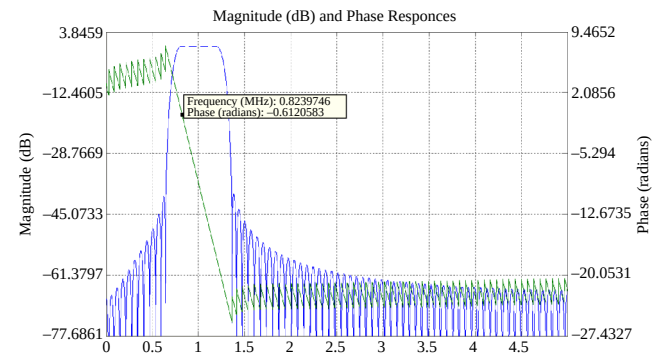
Notes: Kaiser window FIR with order = 153; $F_{p1} = 0.4$ MHz; $F_{p2} = 0.43$ MHz; $F_{s1} = 0.249$ MHz; $F_{s2} = 0.587$ MHz; $R_p = 0.3276$; $R_s = 40.97$; phase delay = -1.159 rad

Figure 13 Band pass filter (harmonic 3)



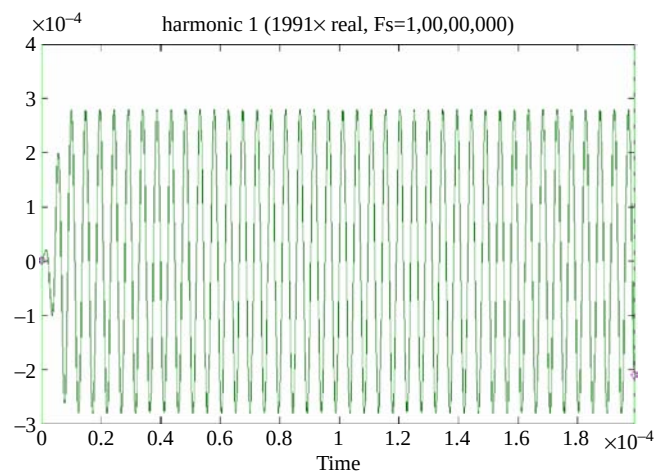
Notes: Kaiser window FIR with order = 162; $F_{p1} = 0.597$ MHz; $F_{p2} = 0.653$ MHz; $F_{s1} = 0.457$ MHz; $F_{s2} = 0.802$ MHz; $R_p = 0.33$; $R_s = 40.29$; phase delay = -0.361 rad

Figure 14 Band pass filter (harmonic 4)



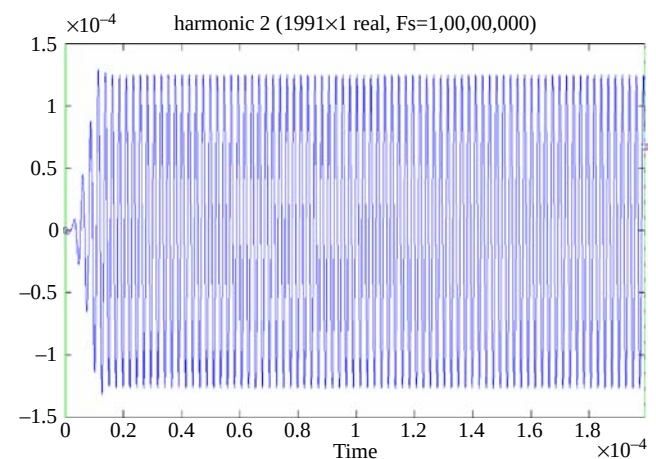
Notes: Kaiser window FIR with order = 148; $F_{p1} = 0.802$ MHz; $F_{p2} = 0.849$ MHz; $F_{s1} = 0.650$ MHz; $F_{s2} = 1.71$ MHz; $R_p = 0.32$; $R_s = 40.25$; phase delay = -0.612 rad

Figure 15 H_1 filtered signal

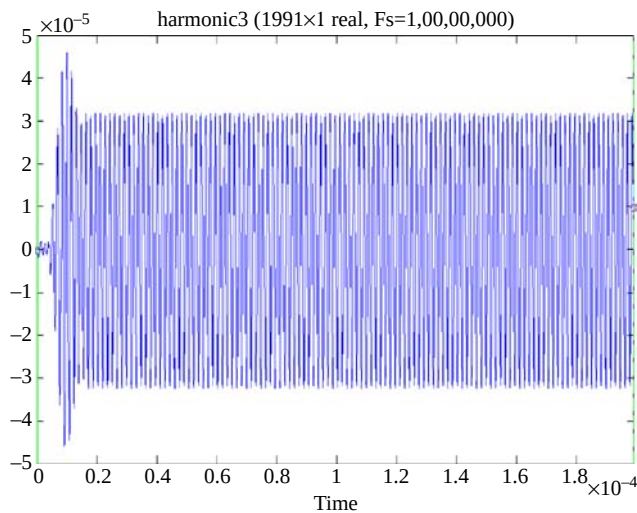


Note: Amplitude = 0.28 mV

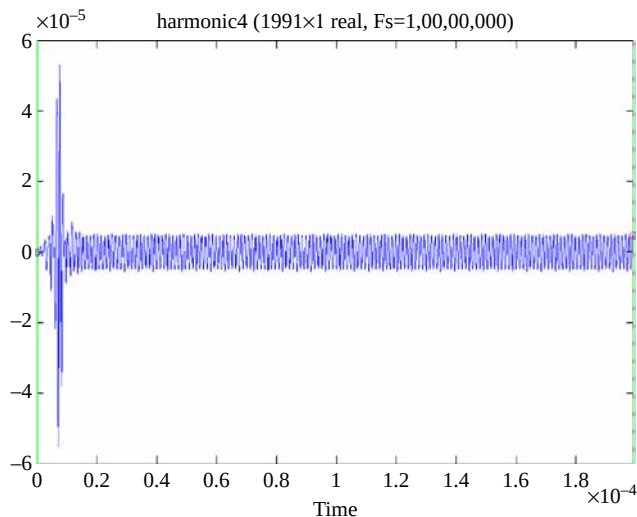
Figure 16 H_2 filtered signal



Note: Amplitude = 0.125 mV

Figure 17 H_3 filtered signal

Note: Amplitude = 0.03 mV

Figure 18 H_4 filtered signal

Note: Amplitude = 0.008 mV

The comparison of Ω calculated here with the input rotation rate, i.e. $10^\circ/\text{s}$ showed a small difference of $0.1^\circ/\text{s}$. This discrepancy caused from utilizing unsharp filters. In experience it was found that, we can get more accuracy by higher order of filters.

VIII. Conclusion

In this paper, the basics of an IFOG were discussed. IFOG output equations were also analyzed to derivate Sagnac phase shift, and a comfort, accurate approach was proposed to calculate the subject phase shift. Finally, the simulation results of the digital signal processing using digital filtering were reported. The results showed that the chosen computational method has reliable accuracy.

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